Mihir Ravindra Adelkar

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Software Engineer with 3+ years building reliable, fault-tolerant distributed systems on AWS and GCP. Reduced MTTR from 25 to 15 min via Prometheus, 45%+ cost reduction through rightsizing, and built Terraform modules adopted across 3 teams. Production engineering in **Python**, **Java**, **C++**, and **Go**.Drawn to systems where scale and reliability matters.

Education

Northeastern University — Boston, MA
Master of Science in Software Engineering Systems — *Sept 2023 – May 2025*

University of Mumbai — Mumbai, India
Bachelor of Engineering in Computer Engineering — *Aug 2018 – Jun 2021*

Technical Skills

Languages: Python, Java, C++, Go, Bash, TypeScript, SQL, Groovy
Cloud & Infra: AWS, GCP, S3, IAM, ALB/NLB, CloudWatch, Terraform, Ansible, Linux, VPC
Containers: Docker, Kubernetes (EKS/GKE), Helm, Istio, mTLS
CI/CD: Jenkins (Pipeline-as-Code, JCasC), GitHub Actions, GitLab CI, Packer, ArgoCD
Observability & Reliability: Prometheus, Grafana, New Relic, OpenTelemetry, ELK Stack, K6

Experience

Full-Stack Developer — Aug 2025 – Present
KeelWorks Foundation — *Remote*

- Own infrastructure for KeelCompass workforce engagement platform on AWS EKS — Terraform modules, GitHub Actions CI/CD, Docker containerization; cut deployment time 60% and release failures 40%
- Set up Prometheus/Grafana dashboards tracking API latency, error rates, and pod health across environments; configured alerting on threshold breaches to surface degradation before user impact
- Automated rollbacks on failed health checks; instrumented Node.js backend services with health probes and structured logging for on-call visibility

Software Engineer — May 2023 – Sep. 2023
Freelancing (Dabadu.ai) — *Remote*

- Debugged flaky production issues across TypeScript, **C++**, and **Python** services in AWS-hosted CRM;
- Added structured logging and wrote runbooks for common incidents (API timeouts, DB connection leaks) referenced by the team during production triage
- Integrated SonarCloud into Bitbucket pipelines; enforced quality gates that reduced post-deploy bugs 20%

Software Engineer — Jun 2021 – May 2023
NeoSOFT Technologies — *Mumbai, India*

- Authored reusable Terraform modules for EKS, RDS, S3, and IAM — adopted as org-wide provisioning standards across 3 engineering teams; managed remote state via S3 backends with workspace-based environment separation
- Built authentication and authorization sidecar in **C++** deployed across 12 microservices, centralizing identity and policy enforcement and eliminating per-service auth duplication
- Coordinated migration of 12 microservices to gRPC + Kafka — 7x faster inter-service calls and 45%+ infrastructure cost reduction via AWS EKS rightsizing
- Cut deploy time from 2 hr to 12 mins via Jenkins blue-green pipelines; ran K6 load tests catching race conditions pre-production; configured EKS HPA and cluster autoscaler handling 3x traffic spikes during product launches
- Set up Prometheus/Grafana dashboards with SLO/SLI tracking for API latency and error rates; alerting cut MTTR from 25 to 15 min; right-sized EKS clusters deferring $30K in hardware spend via capacity planning
- Led blameless postmortems documenting root causes and action items; wrote **Python** and Bash scripts for log rotation, backups, and health checks across 50+ hosts (40% reduction in on-call pages); configured AWS Secrets Manager and least-privilege IAM roles passing security audit with zero critical findings

Projects

CVE Intelligence System — Cloud-Native Infrastructure | *Go, AWS EKS, Terraform, Helm, Istio, Jenkins, Kafka*

- Deployed distributed microservices in **Go** on AWS EKS using Terraform IaC and Helm charts; configured Istio service mesh with mTLS encryption for secure inter-service communication
- Implemented Prometheus/Grafana for real-time observability, alerting, and SLO dashboards; bootstrapped Jenkins with Packer/JCasC for reproducible CI environments

Healthcare Prior Authorization AI System | *Python, FastAPI, Claude API, MCP, FHIR, Pinecone, GCP*

- Engineered AI system replacing 3–5 day manual insurance reviews with automated 5–10 second decisions; built clinical NLP pipeline in **Python** with FastAPI mapping unstructured notes to FHIR resources with 95%+ accuracy
- Designed MCP architecture for context-aware LLM reasoning; system outputs confidence scores with full audit trails for reviewer prioritization